EDS

PRACTICAL NO:02

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CS1-C13

PRACTICE ASSIGNMENT

2.1.1 LIST OPERATIONS

Writea Python programthatimplements a menu-driven interface for managing a list of integers. The program

should havethe followingmenuoptions:

1. Add
2. Remove
3. Display
4. Quit

The program should repeatedly prompt the user to enter a choice from the menu. Depending on the choice selected, the program should perform the following actions:

- Add: Prompts the user to enter an integer and add it to the integer list. If the input is not a valid integer, display "Invalid input".
- Remove: Prompts the user to enter an integer to remove from the list. If the integer is found in the list, remove it; otherwise, display "Element not found". If the list is empty, display "List is empty".
- Display: Displays the current list of integers. If the list is empty, display "List is empty".
- Quit: Exits the program.
- The program should handle invalid menu choices by displaying "Invalid choice".Ensure that the program continues to prompt the user until they choose to quit (option 4).

INPUT

```
I1=[]
while True:
    print("1. Add")
    print("2. Remove")
    print("3. Display")
    print("4. Quit")
    n=int(input("Enter choice: "))
    if n==1:
        add=int(input("Integer: "))
        I1.append(add)
        print(f"List after adding: {I1}")
    elif n==2:
        if len(I1)==0:
            print("List is empty")
        elif len(I1)!=0:
            remove=int(input("Integer: "))
            if remove not in I1:
                print("Element not found")
            else:
                I1.remove(remove)
                print(f"List after removing: {I1}")
    elif n==3:
        if len(I1)==0:
            print("List is empty")
        else:
            print(I1)
    elif n==4:
        break
    else:
        print("Invalid choice")
```

OUTPUT

2.1.2 DICTIONARY OPERATIONS

WriteaPython program toperform insertion, update, deletion, and traversal operations on a dictionary. An initial dictionary containing 10 predefined records is already given in the program.

Operations to be Performed:

1. Insertion – Insert a new key-value pair into the dictionary using user input.
2. Update – Update the value of an existing key using user input.
3. Deletion – Delete a specified key from the dictionary using user input.
4. Traversal – Traverse the final dictionary and display all key-value pairs.

Input and Output Format:

1. Read an integer representing the key to be inserted and a string representing its value. Insert this new key-value pair into the dictionary and display the dictionary after insertion.
2. Read an integer representing the key to be updated and a string representing the new value. Update the value of the specified key (only if it exists) and display the dictionary after the update.
3. Read an integer representing the key to be deleted. Delete the specified key from the dictionary (only if it exists) and display the dictionary after deletion.
4. Finally, traverse the dictionary and display all key-value pairs.

The program should also display the original dictionary before performing any operations. Each output should be printed with appropriate messages.

Note:

- All operations must be performed using dictionary methods.
- Perform deletion only if possible; leave the dictionary unchanged.
- Refer to the visible test cases for better understanding and strictly match with the input/outputs.

INPUT

```
# Initial dictionary with 10 predefined records
student = {
    1: "Amit",
    2: "Riya",
    3: "Kiran",
    4: "Neha",
    5: "Arjun",
    6: "Pooja",
    7: "Rahul",
    8: "Sneha",
    9: "Vikram",
    10: "Anjali"
}

# write your code here...

print("Original Dictionary:",student)
#Step 1
key1=int(input())
value1=input()
student.update({key1:value1})
print("After Insertion:",student)
#Step 2
key2=int(input())
value2=input()
if key2 in student.keys():
    student.update({key2:value2})
print("After Update:",student)
#Step 3
key3=int(input())
if (key3 in student.keys()):
    student.pop(key3)
print("After Deletion:",student)
#Step 4
print("Traversing Dictionary:")
for key,value in student.items():
    print(f"{key} : {value}")
```

OUTPUT

Average time: 0.040 s / 39.50 ms

Maximum time: 0.045 s / 45.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 45 ms Debug

Expected output	Actual output
Original Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}	Original Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}
11	11
Suresh	Suresh
After Insertion: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}	After Insertion: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}
3	3
Karthik	Karthik
After Update: {1: 'Amit', 2: 'Riya', 3: 'Karthik', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}	After Update: {1: 'Amit', 2: 'Riya', 3: 'Karthik', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}
5	5
After Deletion: {1: 'Amit', 2: 'Riya', 3: 'Karthik', 4: 'Neha', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}	After Deletion: {1: 'Amit', 2: 'Riya', 3: 'Karthik', 4: 'Neha', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}
Traversing Dictionary:	Traversing Dictionary:
1 : Amit	1 : Amit
2 : Riya	2 : Riya
3 : Karthik	3 : Karthik
4 : Neha	4 : Neha
6 : Pooja	6 : Pooja
7 : Rahul	7 : Rahul
8 : Sneha	8 : Sneha
9 : Vikram	9 : Vikram

Test case 2 40 ms Debug

Expected output	Actual output
Original Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}	Original Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}
12	12
Meera	Meera
After Insertion: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 12: 'Meera'}	After Insertion: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 12: 'Meera'}
6	6
Divya	Divya
After Update: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Divya', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 12: 'Meera'}	After Update: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Divya', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 12: 'Meera'}
1	1
After Deletion: {2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Divya', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 12: 'Meera'}	After Deletion: {2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Divya', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 12: 'Meera'}
Traversing Dictionary:	Traversing Dictionary:
2 : Riya	2 : Riya
3 : Kiran	3 : Kiran
4 : Neha	4 : Neha
5 : Arjun	5 : Arjun
6 : Divya	6 : Divya
7 : Rahul	7 : Rahul

SELF ASSIGNMENT

2.2.1 LINEAR SEARCH TECHNIQUE

Write aprogramtocheckwhetherthegivenelementispresentornotinthearrayofelementsusinglinearsearch.

Input format:

- The first line of input contains the array of integers which are separated by space
- The last line of input contains the key element to be searched

Output format:

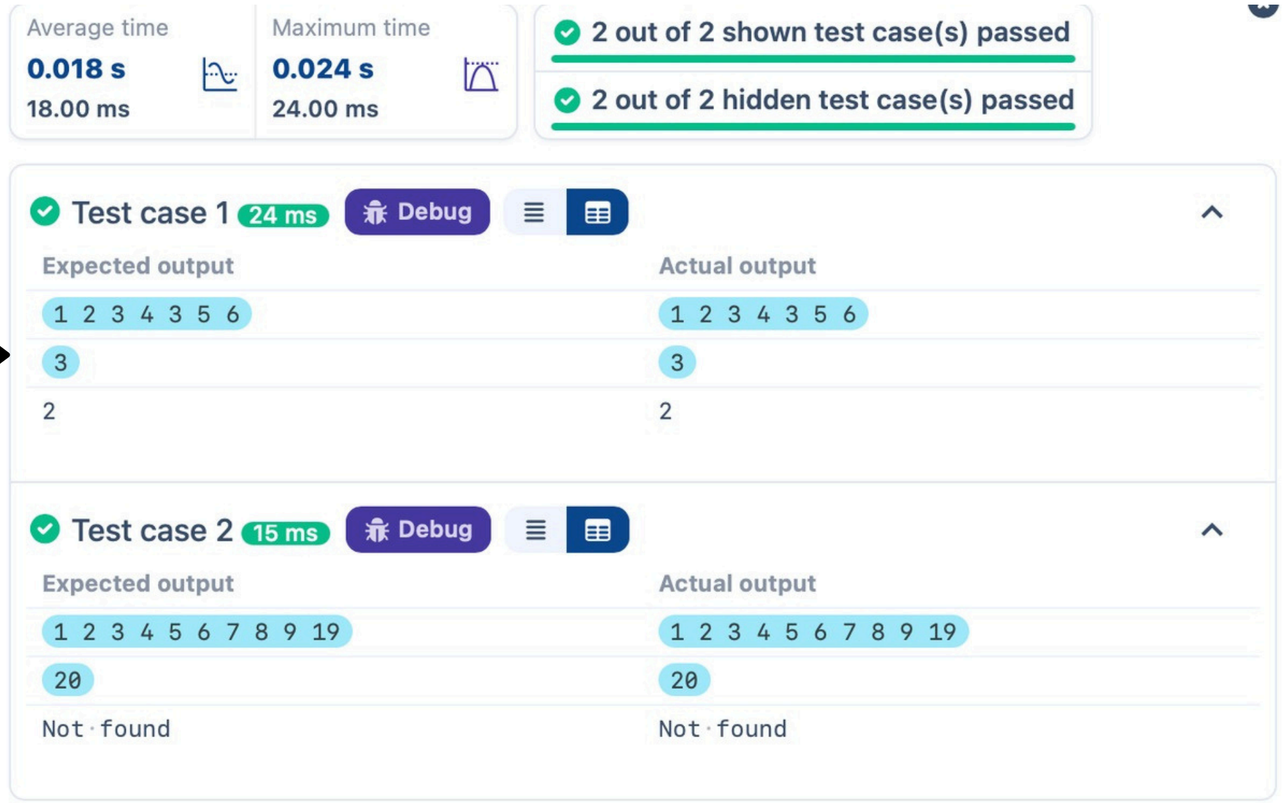
- If the element is found, print the index. If the element found multiple times, print the starting index
- If the element is not found, print Not found.

Sample Test Case:
Input:
1 2 3 4 3 5 6
3
Output:
2

INPUT

```
def linear_search(arr,key):
    for i in range(len(arr)):
        if arr[i]==key:
            return i
    return -1

arr=list(map(int,input().split()))

key=int(input())

result=linear_search(arr,key)

if result !=-1:
    print(result)

else:
    print("Not found")
```

OUTPUT



2.1.2 CAPTAIN OF THE TEAM

You are providedwith the heights of 11cricketplayers(in centimeters). Your task is to identify the tallest player, who will beselectedas the captain of theteam.

Input Format:
The first line of input will contain 11 integers, each representing the height of a player (in centimeters), each separated by a space.

Output Format
The output should be the height (in centimeters) of the tallest player.

INPUT

```
heights= list(map(int,input().split()))
tallest_player=max(heights)
print(tallest_player)
```

OUTPUT